

old	AAATTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTAACAGGAAG	AGCTTGCTTCTTTGCTGACGAGTGGCGGAC	110
new	AGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTAACAGGAAG	CAGCTTGCTGCTTTGCTGACGAGTGGCGGAC	101
consensus	!!	!!	
old	GGGTGAGTAATGTCTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTAGCTAATACCGCATAACGTCGCAAG	ACCAAAGAGGGGGACCTTCGGGCCTCTTG	220
new	GGGTGAGTAATGTCTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTAGCTAATACCGCATAACGTCGCAAG	CACAAAGAGGGGGACCTTAGGGCCTCTTG	211
consensus	!!	!!	
old	CCATCGGATGTGCCCAGATGGGATTAGCTAGTAGGTGGGGTAACGGCTCACCTAGGCGACGATCCCTAGCTGGTCTGAGAGGATGACCAGC	CACACTGGAAGTGAACAC	330
new	CCATCGGATGTGCCCAGATGGGATTAGCTAGTAGGTGGGGTAACGGCTCACCTAGGCGACGATCCCTAGCTGGTCTGAGAGGATGACCAGC	ACACTGGAAGTGAACAC	321
consensus	!!	!!	
old	GGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGC	C GCGTGTATGAAGAAGGCCTTCGGGTTGTAAAGTACTTTC	440
new	GGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGC	N GCGTGTATGAAGAAGGCCTTCGGGTTGTAAAGTACTTTC	431
consensus	!!	!!	
old	AGCGGGGAGGAAGGGAGTAAAGTTAATACCTTTGCTCATTGACGTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGTGCAAGCG		550
new	AGCGGGGAGGAAGGGAGTAAAGTTAATACCTTTGCTCATTGACGTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGTGCAAGCG		541
consensus	!!		
old	TTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTTTGTTAAGTCAGATGTGAAATCCCCGGGCTCAACCTGGGAAGTGCATCTGATACTGGCAAGCTTGAGTCTC		660
new	TTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTTTGTTAAGTCAGATGTGAAATCCCCGGGCTCAACCTGGGAAGTGCATCTGATACTGGCAAGCTTGAGTCTC		651
consensus	!!		
old	GTAGAGGGGGGTAGAATTCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGTGGCGAAGGCGGCCCCCTGGACGAAGACTGACGCTCAGGTGCGAAAGC		770
new	GTAGAGGGGGGTAGAATTCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGTGGCGAAGGCGGCCCCCTGGACGAAGACTGACGCTCAGGTGCGAAAGC		761
consensus	!!		
old	GTGGGGAGCAAACAGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCGACTTGGAGGTTGTGCCCTTGAGGCGTGGCTTCCGGA	GC TAACGCGTTAAGTCGACC	880
new	GTGGGGAGCAAACAGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCGACTTGGAGGTTGTGCCCTTGAGGCGTGGCTTCCGGA	NN TAACGCGTTAAGTCGACC	871
consensus	!!	!!	
old	GCCTGGGGAGTACGGCCGCAAGGTTAAACTCAAATGAATTGACGGGGGCC	GCACAAGCGGTGGAGCATGTGGTTTAAATTCGATGCAACGCGAAGAACCTTACCTGGTC	990
new	GCCTGGGGAGTACGGCCGCAAGGTTAAACTCAAATGAATTGACGGGGGCC	GCACAAGCGGTGGAGCATGTGGTTTAAATTCGATGCAACGCGAAGAACCTTACCTGGTC	980
consensus	!!	!!	
old	TTGACATCCACGGAAGTTTTTCAAGATGAGAATGTGCCTTCGGGAACCGTGAGACAGGTGCTGCATGGCTGTCGTGAGCTCGTGTGTGAAATGTTGGGTAAAGTCCCGC		1100
new	TTGACATCCACGGAAGTTTTTCAAGATGAGAATGTGCCTTCGGGAACCGTGAGACAGGTGCTGCATGGCTGTCGTGAGCTCGTGTGTGAAATGTTGGGTAAAGTCCCGC		1090
consensus	!!		
old	AACGAGCGCAACCCTTATCCTTTGTTGCCAGCGGTCCGGCCGGGAACCTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATCATGGCCC		1210
new	AACGAGCGCAACCCTTATCCTTTGTTGCCAGCGGTCCGGCCGGGAACCTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATCATGGCCC		1200
consensus	!!		

old

new

consensus

TTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACCTCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTCGTAGTCCGGATTGGAGTCTGCAAC
TTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACCTCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTCGTAGTCCGGATTGGAGTCTGCAAC
!!

1320
1310

old

new

consensus

TCGACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCCCGGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAA
TCGACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCCCGGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAA
!!

1430
1420

old

new

consensus

AGAAGTAGGTAGCTTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAAGGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCT
AGAAGTAGGTAGCTTAAC.TTCGGGAGGGCG.....
!!!!!!!!!!!!!!!!!!

1540
1450

old

new

consensus

TA 1542
.. 1450

⧻

⧻

non-conserved
≥ 50% conserved